

A Risk-Sensitive Learning-to-Defer Framework for Responsible AI in Financial Fraud Detection

Project Report

MSc Research Project/ (Research) Practicum/ Internship

MSc in Artificial Intelligence for Business

Dhwani Sanjay Kariya

School of Computing
National College of Ireland

Supervisor: Dr Muslim Jameel Syed

Date: 10/08/2026

A Risk-Sensitive Learning-to-Defer Framework for Responsible AI in Financial Fraud Detection

Dhwani Sanjay Kariya

Abstract

The automated fraud detection system should route the most high-risk cases to a human but all published Learning-to-Defer (L2D) models consider a misclassified €500 transaction equally expensive to defer as a misclassified €500,000 transaction, which is in contradiction with Article 14 of the EU AI Act, which says supervision should be in accordance with the risk involved. This thesis looks at the ability of a risk-sensitive weighting factor used in the L2D deferral objective to improve the safety of the most high-stakes cases, by using an instance-level cost instead of a fixed deferral cost. The instance-level cost is determined by a four-tier risk matrix derived from the distribution of transaction amounts in each dataset. The three models are trained on ULB Credit Card Fraud and tested on IEEE-CIS and PaySim. The Standard L2D is not risk-proportional on any of the datasets but rather is inverted on IEEE-CIS, with the lower-value transactions (12.74%) having a higher deferral rate and the higher-value transactions (10.67%) having a lower deferral rate. Risk-sensitive weighting addresses this. Critical-tier deferral is increased by a factor of 4.6x, 6.6x and 5.65x on ULB, IEEE-CIS and PaySim, respectively, compared to the standard L2D. The effect requires a fairly large cost difference and a rule-based baseline across the three datasets indicates that the method is consistently better at being efficient. The contribution is the loss modification that is independent of the architecture and is validated using multi-seed robustness checks and bootstrap confidence intervals.

1. Introduction

Most organisations have assumed that there will always be a human review to cover for the mistakes made by AI when processing loan applications, insurance claims and financial transactions. This safety net, in practice, is less effective than assumed, as people are more willing to accept an AI's output even in the presence of errors, a phenomenon known as automation bias, both in laboratory and in deployed systems (Bansal *et al.*, 2021; Buçinca, Malaya and Gajos, 2021). If the model can't be designed to flag the most risky decisions to be made by humans, then the model is not meant to be used for human judgment.

One technical solution is to learn to Defer (L2D), a model that learns the correct classification as well as whether a case should be deferred to a human expert or not. The first statistically consistent formulation was provided by (Mozannar and Sontag, 2021), later followed by fully differentiable formulations (Okati, De and Gomez-Rodriguez, 2021; Hemmer *et al.*, 2023; Mozannar *et al.*, 2023) which cut the expert-labelled data required and replaced the approximations with exact algorithms. The EU AI Act ("Regulation (EU) 2024/1689 of the European Parliament and of the Council of 13 June 2024 laying down harmonised rules on

artificial intelligence (Artificial Intelligence Act),” 2024) introduces the requirement for human oversight of high-risk AI systems such as financial services but there is no published L2D method to comply with this requirement.

All published L2D formulations have one drawback - they all treat all misclassified decisions equally, meaning that a misclassified €500 transaction is as bad as a misclassified €500,000 transaction. This is not a bug, it is by design and it's never been tested.

Research Question:

What happens to the rate of unsafe automation in high-stakes decision systems when a risk-sensitive weighting factor is added to a Learning-to-Defer model?

Hypothesis: Replacing the uniform deferral cost with an instance-level, risk-weighted cost will (H1) increase the deferral rate on Critical-tier transactions relative to standard L2D and (H2) reduce the number of high-value frauds missed, without a fixed or predictable cost to overall classification accuracy.

Research Objectives:

1. Construct a data-driven, percentile-based risk matrix for each dataset, independent of any model training, to avoid post-hoc tuning of the risk definition.
2. Implement three models - an automation-only baseline, a standard fixed-cost L2D model and a risk-weighted L2D model - that are identical in every respect except the deferral cost term, to isolate the causal effect of risk-sensitive weighting.
3. Evaluate whether risk-sensitive weighting increases deferral in proportion to financial risk across three structurally different fraud datasets (ULB, IEEE-CIS, PaySim).
4. Test the robustness of the effect against random seed variation, risk-weight configuration and a simple rule-based alternative, to distinguish a genuine effect from an artefact of one lucky run.

This method proposed here uses an instance-level cost that changes according to the risk of the transactions. Each dataset's actual distribution of transaction amounts is used to create a four-tier risk matrix (Low, Medium, High, Critical) and not on arbitrary thresholds. The higher tiers have lower deferral cost and encourage the model to defer precisely those that would be most important if they were wrong.

Three contributions are made in this thesis: technically, the first L2D formulation to weight cost directly in the deferral objective, instead of assuming it to be fixed; empirically, a comparison of risk-sensitive to standard L2D, on three datasets of public fraud; practically, meeting the Article 14 oversight requirement by learning the deferral threshold during training, rather than applying a review policy after deployment.

The remainder of this report is organised as follows: Section 2 summarises the concepts of automation bias, the history of L2D and the regulatory context; Section 3 summarises the research methodology; Section 4 discusses the design of the three models and the risk-sensitive loss function; Section 5 discusses implementation; Section 6 discusses and summarises results and Section 7 concludes and outlines future work.

2. Related Work

2.1 The Persistent Failure of Passive Safeguards

This is a growing trend of using AI as a component of human-AI teams (Mosqueira-Rey *et al.*, 2023) and Learning to Defer (L2D, Section 2.2) falls in this picture but many papers in the human-AI literature assume that the human is already closely monitoring. In reality, this does not happen, AI does not support the judgment of the individual but it takes its place. Explainable AI research (Arrieta *et al.*, 2020) hypothesized that a decision's rationale would lead to increased scrutiny of the decision. (Bansal *et al.*, 2021) conducted a controlled study on 1,626 participants and obtained the opposite results: explanations did not lead to any complementary team performance increase when used alongside humans (or alongside AI alone) and they did lead to increases in relying on incorrect predictions just when they were needed the most.

(Buçinca, Malaya and Gajos, 2021) applied cognitive forcing functions, interface designs that require that users take action instead of passively accept the design. In total, they managed to decrease over-reliance in 199 participants but the top three performing designs were the least liked by users and even the top one did not overcome the performance of the AI itself. While interface changes may reduce but not eradicate unsafe AI uses and changes that are most effective are those that are resisted by users. Where the human side can't be dealt with reliably, the more pragmatic way is to alter the system itself.

2.2 Learning to Defer: From Theoretical Foundations to Practical Constraints

Learning to Defer (L2D) is a learning system that is not forecasting but rather passing a case to a human expert. The problem was introduced by (Madras, Pitassi and Zemel, 2018) as a fairness problem. (Mozannar and Sontag, 2020) showed that a rejection-based model trained by a standard surrogate loss does not converge to the true defer/predict boundary and proposed that the remedy to this issue is to train both models together, as is done in Model 2 here. This was later repeated to achieve full differentiability, to reduce the number of data labelled by experts and to increase the number of experts (Keswani, Lease and Kenthapadi, 2021; Okati, De and Gomez-Rodriguez, 2021; Hemmer *et al.*, 2023). Most importantly, (Liu, Gallego and Barbieri, 2022) consider model uncertainty as a source of deferral, although model uncertainty can be low and deferral can be high.

In their previous work, (Mozannar *et al.*, 2023) demonstrated that such approximation approaches result in inaccurate boundaries and proposed exact algorithms. Selective classification, which is related but different from L2D, is one in which a model does not make any decision; it does not assign the case to someone else, nor does it consider the costs of not taking action (Pugnana and Ruggieri, 2023; Xia and Bouganis, 2024). However, L2D has come from a concept to near-deployment in five years and all of the versions take for granted that the cost of error is the same no matter what harm is done (which is not the case for plain classification - see (Elkan, 2001; Bahnsen, Aouada and Ottersten, 2015))

2.3 Benchmarking L2D: Progress Confirmed, a Risk Left Unexamined

(Alves *et al.*, 2025) published OpenL2D as a benchmark in Scientific Data for the first time. It demonstrates that state-of-the-art L2D results in significantly less unsafe automation than a plain classifier and suggests simulating expert annotations by using an instance-dependent noise (IDN) model when there is no real label. This thesis is simpler since, as there are no real expert labels for the three datasets here, the deferral head is compared to the ground-truth label. If there is a great variation in the costs of error, as this thesis tests, a model learned from a single cost should systematically under-defer when the true error cost is high.

2.4 The Governance Imperative: A Technical Bridge That Does Not Yet Exist

Governance research takes an alternate approach to the conclusion that accuracy is not sufficient. Indeed, the (High-Level Expert Group on Artificial Intelligence, 2019) previously concluded that trustworthy AI systems can only be under human control. (Raji *et al.*, 2020) conducted a survey of the state of AI accountability in practice and found that it was largely absent with no review process or audit trail. (Mökander and Floridi, 2023) replied with an ethics-based audit which is possible but challenging to maintain over time. Raji *et al.* determine that the place of accountability is missing, (Mökander and Floridi, 2023) find ways to detect accountability failures once they have occurred; both do not prevent a high-risk decision from being taken in the first place.

(Buscemi *et al.*, 2025) studied the EU AI Act with respect to high-risk AI systems, specifically (“Regulation (EU) 2024/1689 of the European Parliament and of the Council of 13 June 2024 laying down harmonised rules on artificial intelligence (Artificial Intelligence Act),” 2024), which calls for human oversight of high-risk financial systems but does not explain how. (Sterz *et al.*, 2024) say that having an overwhelming amount of routine cases for review is not true oversight, which is a direct case for a risk-proportional queue for review, a mechanism that a risk-sensitive L2D model builds into itself and not as an additional policy.

2.5 Research Niche and Expected Contribution

Three gaps have emerged in the above research. When the interface design offers solutions, people over-rely on AI in ways that can't be addressed by interface design (Bansal *et al.*, 2021; Buçinca, Malaya and Gajos, 2021). Until now, no L2D formulation across several generations, have considered the real-world cost of an error (Madras, Pitassi and Zemel, 2018; Keswani, Lease and Kenthapadi, 2021; Mozannar and Sontag, 2020; Okati, De and Gomez-Rodriguez, 2021; Liu, Gallego and Barbieri, 2022; Hemmer *et al.*, 2023; Mozannar *et al.*, 2023; Alves *et al.*, 2025). To address this, a cost-sensitive learning method corrected the issue for plain classification (Elkan, 2001; Bahnsen, Aouada and Ottersten, 2015) but not for deferral. The governance research has dealt with the accountability issue, but could not give any technical fix (High-Level Expert Group on Artificial Intelligence, 2019; Raji *et al.*, 2020; Mökander and Floridi, 2023; Sterz *et al.*, 2024; Buscemi *et al.*, 2025). The gaps here are not discrete, in the case of automation bias, it made the human backstop not as reliable when it is needed the most (2.1) but in the case of cost-sensitive learning, it is already possible to do the weighting of errors by consequences (2.2) but there is no mechanism to reach the proportional oversight that regulators desire (2.4). This thesis tackles all three through a lever, a lever of a risk-sensitive weighting factor in the L2D training objective, which is evaluated on three public datasets of fraud-detection problems, in the context of the EU AI Act's oversight requirement.

3. Research Methodology

3.1 Research Method

It is a quantitative experimental study: three models are trained on the same data and the same set of conditions with only the cost weighting in the deferral objective different. The base deferral cost is the same for both models 2 and 3 of a specific dataset, the only difference is that the base deferral cost is applied at the case level or at the base level. The upper bound for unsafe automation is provided by a simple classifier, Model 1, which does not delay the judgment. Model 2 is standard L2D according to (Mozannar and Sontag, 2021) using a single

cost. Model 3 changes this fixed cost with the weight associated to each instance, as specified in Section 4.

All three datasets are without any real expert annotations, following a simplification applied in other parts of L2D literature (Mozannar and Sontag, 2020; Alves *et al.*, 2025) where the deferral head is trained directly on the joint loss with the ground-truth fraud label. It's a reasonable question to ask, if we don't have any expert-comparison term, does this now become a Learning-to-Defer, or rather a cost-sensitive reject-option classifier? It's an architectural difference. Selective classification (Pugnana and Ruggieri, 2023; Xia and Bouganis, 2024) considers abstention as a characteristic of the classifier without any target. The trick here is to train a second head not post-hoc with a confidence but together with the classifier on a shared trunk; the two heads are learned from the predicted difficulty of the case and the cost of the instance. The idea here is to learn a second head jointly with the classifier on a shared trunk, instead of removing the L2D architecture and replacing the ground truth with a simulated expert (i.e., learned from the predicted difficulty of the case and the cost of the instance). The deferral head is trained jointly with the classifier on a shared trunk, learning from the predicted difficulty of the case and the instance cost, rather than being fitted after training on classifier confidence the way selective classification is.

3.2 Research Resources

Three publicly available datasets are used: ULB Credit Card Fraud (Dal Dal Pozzolo *et al.*, 2015), primary), IEEE-CIS Fraud Detection (IEEE-CIS, 2019, validation) and PaySim Mobile Money Simulation (Lopez-Rojas, Elmir and Axelsson, 2016), validation). Dataset-specific preprocessing is covered in Section 5.

3.3 Evaluation Protocol

Five numbers are reported for each model on each dataset: accuracy and F1 of the autonomous decisions (the former is a misleading metric due to the severity of the class imbalance), ROC-AUC of the autonomous decisions, overall deferral rate and deferral rate split by risk tier, the main interest of this thesis, because the question concerns the increase in the deferral rate rather than whether it increases. Fraud level counts (caught autonomously, missed, deferred) are also provided to help make the abstract metrics real. No model training and evaluation is done before the construction of the risk matrix, so the risk matrix is not adjusted after the training and evaluation for the purpose of beautifying the results.

3.4 Ethical Considerations

No human subjects are involved in this study. PaySim is a fully synthetic data set, while ULB and IEEE-CIS are anonymised (PCA-transformed).

There are two issues that need to be addressed from a practical perspective. Class imbalance: the training set is only augmented with SMOTE and the metrics are only calculated per class and risk tier, not one accuracy figure, since training without adjustment would optimise mostly for the majority class. Second, the precondition of the framework, risk-sensitive deferral is only effective if the organisation has sufficient reviewers to process the additional volume it will send to the human reviewers, otherwise it becomes a "backlog" and not a safety improvement, which requires the framework to be coupled with a proper capacity assessment of reviewers before it can be used in practice.

4. Design Specification

4.1 Risk Matrix Design Philosophy

The risk matrix is the key to making financial risk usable in a loss function and it does not have any arbitrary fixed thresholds. The transaction amount distributions of each dataset create a matrix of percentiles (Table 1). A percentile matrix asks the same question about each set of data, no matter the scale: Is this transaction unusually large for this population? That enables the same four-tier logic to be applied to IEEE-CIS and PaySim without having to redefine risk every time. All three datasets share the same percentile cut points (25th/50th/75th) but only the currency values are different, as the underlying distributions are different.

Table 1: Percentile-based risk matrix thresholds, ULB (primary dataset).

Tier	ULB (primary)	Basis
Low	€0 - €22	at/below median
Medium	€22 - €77	median - 75th percentile
High	€77 - €500	75th percentile - ~99th percentile
Critical	above €500	top ~1% of transaction values

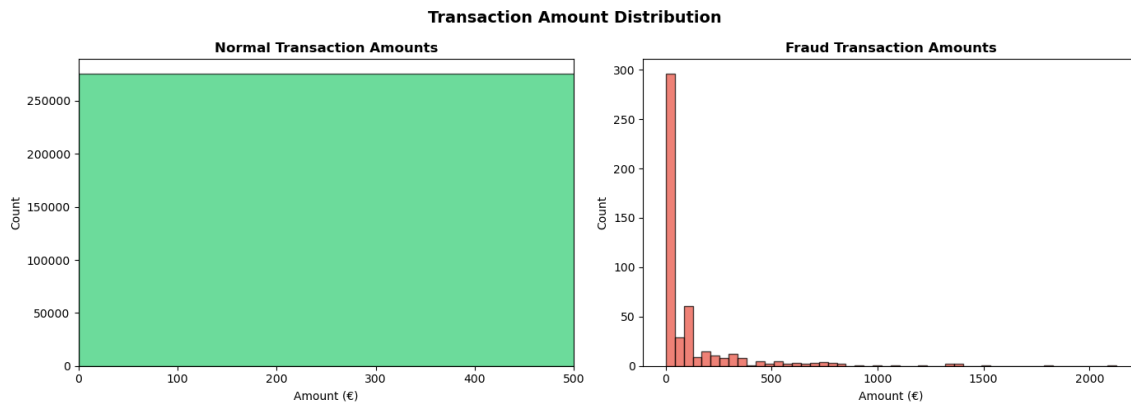


Figure 1: Transaction amount distribution for normal vs fraudulent transactions (ULB). Fraud amounts skew heavily towards the lower end, which is why a percentile-based risk matrix is needed rather than fixed currency bands.

4.2 Empirical Justification for Risk-Sensitive Weighting (ULB)

The raw data is enough to justify that treating tiers differently is quite important, prior to training any model (Table 2). The Critical-tier accounts for 9,142 of the 284,807 ULB transactions (3%) and by far has the highest total fraud value of the four tiers, which is €31,195 compared to the €808 total fraud value of the Low-tier. Tiers are not perfectly monotonic, Low has a higher fraud rate (0.190%) than Medium (0.066%), so size is not the only factor that tracks the fraud rate. Financial exposure increases with tier. The risk matrix is based on the expected loss per transaction which increases from €0.006 (Low) to €0.028 (Medium), €0.42 (High), €3.41 (Critical). Here, the deferral cost is tiered and not uniform because the Critical-tier is also the most heavily frauded tier in all the three datasets.

Table 2: Transaction counts, fraud counts, fraud rate, total fraud value and average amount by risk tier (ULB), establishing the empirical case for risk-sensitive weighting.

Risk Tier	Transactions	Frauds	Fraud Rate	Total Fraud Value	Avg Amount
Low	142,643	271	0.190%	€808	€7
Medium	70,898	47	0.066%	€2,006	€44

Risk Tier	Transactions	Frauds	Fraud Rate	Total Fraud Value	Avg Amount
High	62,124	139	0.224%	€26,118	€183
Critical	9,142	35	0.383%	€31,195	€1,052

4.3 The Three Models

Model 1: automation-only baseline: a simple classifier without deferral, an upper bound on unsafe automation. This is a scikit-learn LogisticRegression on ULB and the same shared-layer network below with an extreme uniform cost deactivated (because a linear model is not fair on 209 features (IEEE-CIS) or 7 (PaySim)). In Model 2, standard L2D, a wrong autonomous choice is punished, irrespective of its value and the cost term is a fixed scalar, not an expert-correctness comparison (Section 3.1). Model 3, risk-sensitive L2D, is the same as Model 2 but the uniform cost is replaced with a term at an instance level:

$$\text{instance_cost}(x_i) = \text{risk_weight}(\text{tier}(x_i)) * \text{base_cost}$$

with risk weights Low = 1.0, Medium = 0.8, High = 0.5, Critical = 0.2 for ULB. The model is much more likely to pass on a high-value case when deferring a Critical-tier misclassification compared to a Low-tier one, at a cost of five times less. The only difference between the two is the variable, the other variables are all the same: architecture, optimiser, epochs, base cost, split and SMOTE.

4.4 Network Architecture

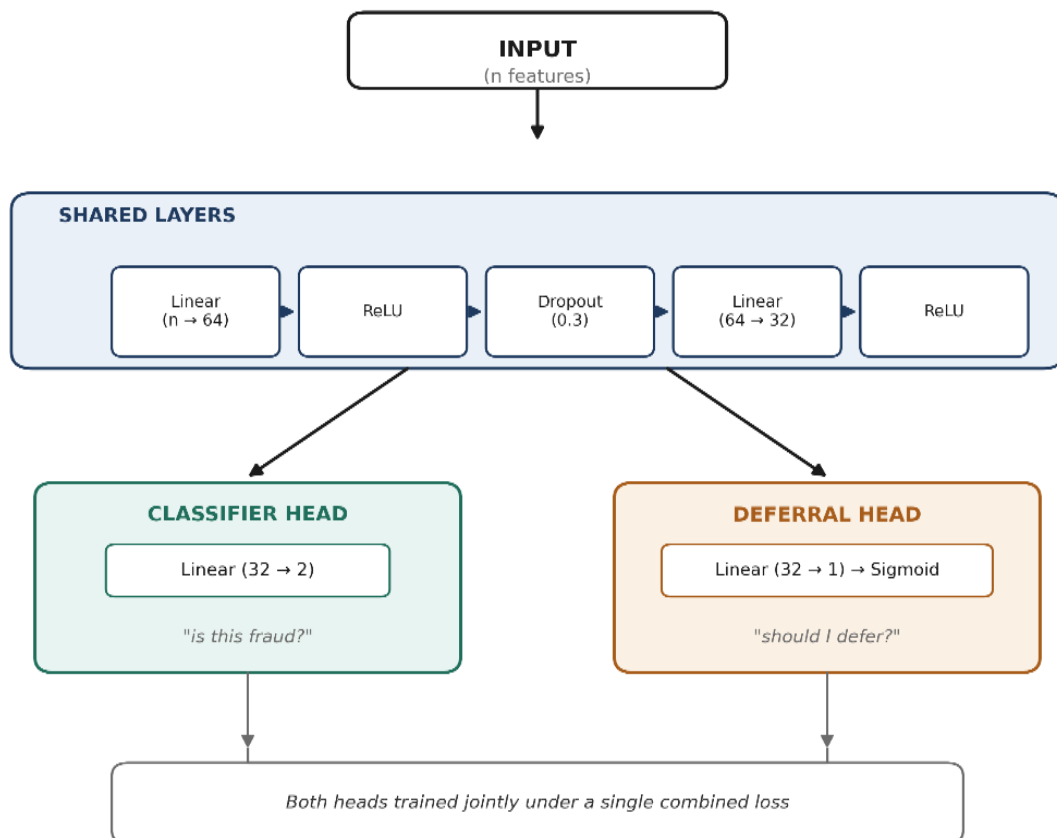


Figure 2: Classifier and deferral heads trained jointly - shared-trunk network architecture.

$n = 30$ for ULB, 209 for IEEE-CIS (dropping features with more than 50% missing values and keeping numeric columns only) and 7 for PaySim (amount, four balance fields, transaction step and a binary transaction-type flag).

4.5 Loss Functions

For higher-risk tiers, the transactions are made not so costly to defer as $w(\text{tier})$ is smaller, so that the learned deferral probability is higher for the most serious transactions according to the risk matrix. This is a direct example of cost-sensitive risk minimisation (Elkan, 2001), the objective is now a weighted expected risk, $E[w(\text{tier}(x)) \cdot \text{loss}(x)]$, with standard L2D being a special case ($w(\text{tier}) = 1$ for all tiers); the gradient contribution of a Critical-tier deferral is scaled down, pushing the optimiser towards a boundary that defers disproportionately on the cases the risk matrix labels as most consequential.

Standard (uniform-cost) L2D loss, per batch:

$$L = (1 - p_{\text{defer}}) * \text{CrossEntropy}(\text{class_logits}, y) + p_{\text{defer}} * c$$

Risk-Sensitive L2D loss. The only change is that the scalar c becomes a per-instance value:

$$L = (1 - p_{\text{defer}}) * \text{CrossEntropy}(\text{class_logits}, y) + p_{\text{defer}} * (w(\text{tier}_i) * c_{\text{base}})$$

4.6 Deferral Cost Calibration

The c_{base} is not set as a post-hoc parameter but rather is tuned per dataset by a small grid search for the value that maintains a workable range of overall deferral (not 0% or 100%). The c_{base} is the same for Model 2 and Model 3, the only difference is the risk-weight multiplier. Final values are - ULB $c_{\text{base}} = 0.13$ (weights Low 1.0 / Medium 0.8 / High 0.5 / Critical 0.2) because the base fraud rate in IEEE-CIS is already high, it does not adopt a very aggressive scheme. In this feature-poor (7-feature) dataset, the response to cost is bimodal, not smooth and costs from 0.13 to 1.0 all converged to around 0% or 99.6% deferral except when $c_{\text{base}} = 0.4$, which ended up in a more stable middle ground (0.45% for Model 2, 1.28% for Model 3) over a complete 15-epoch run and that value was used for all of the PaySim results (Sections 6.3, 6.5.2). Here the value of c_{base} was selected using only the training-set deferral rate as the sole criteria before finding any test-set metric and so the headline results reported in Section 6 are not circular with respect to this choice.

5. Implementation

5.1 Tools and Environment

Implements in Python 3 (Anaconda), PyTorch for the L2D network, custom loss functions, scikit-learn for the ULB baseline network and preprocessing, pandas/NumPy for data handling and Matplotlib/Seaborn for result charts. It can be easily trained on a standard CPU, as no GPU was required for all three datasets.

5.2 Reproducibility

All operations that involve randomness, such as train/test split, SMOTE, weight initialisation and mini-batch shuffling, are seeded (using `random_state = 42` for scikit-learn and imbalanced-

learn, torch.manual_seed(42) as well as a seeded DataLoader generator for PyTorch). To demonstrate the variation this randomness can create, Section 6 provides a robustness check on ULB for different seed values for the data split.

5.3 Data Preparation

The same steps are followed for all datasets, first loading the raw data, then creating the risk tier column by using percentile thresholds specific to the dataset (as outlined in Section 4.1), then scaling continuous features with StandardScaler, then splitting into training and test sets (80/20, stratified, random_state = 42) and finally using SMOTE (Chawla *et al.*, 2002) on the training set only, interpolating between fraud cases and their closest neighbours rather than copying them. The natural, unbalanced distribution of the test set is maintained.

Handling of the data set: ULB scales the data set independently and uses the 28 PCA-transformed features (V1-V28) as they are. IEEE-CIS combines the transaction and identity tables on TransactionID, removes columns that have more than 50% missing values and fills in the missing values with the median for the remaining ones, then scales the result into a 209-feature matrix. PaySim only stores the amount and 4 balance fields, time step and a transaction-type flag; the remaining 7 fields are kept, as fraud is never possible in the other transactions.

5.4 Model Training

All three models have similar architecture in the trunk-plus-two-heads model and were trained for 15 epochs with Adam (learning rate 0.001), batch size 256 (ULB, IEEE-CIS) or 512 (PaySim, to maintain the number of batches similar to the other two models as the number of samples after SMOTE was larger). The base deferral cost c_{base} is tuned once per dataset by grid search and common to Model 2 and Model 3, the only difference being the weighting scheme. The values of the parameters for the final model are: $c_{\text{base}} = 0.13$ (ULB), 0.63 (IEEE-CIS), 0.4 (PaySim, obtained by a special calibration sweep, Section 4.6).

5.5 Outputs Produced

The three trained model artefacts (a fitted scikit-learn estimator for ULB Model 1) with three result JSONs per model are generated for accuracy, F1, ROC-AUC, deferral rate, fraud-level breakdown and deferral rate by risk tier and the comparison figures used in the Evaluation chapter are also generated and stored.

The outputs generated and the additional checks described in Section 3.4 are also in the same format - a JSON result for each weight configuration for the ULB sensitivity sweep, a JSON result for each dataset for the rule-based baseline and a JSON result for each additional seed for the multi-seed checks, in addition to the comparison figures presented in Sections 6.1.2, 6.2.2 and 6.3.2.

6. Evaluation

This chapter shows the results for ULB Credit Card Fraud as the main dataset and results for IEEE-CIS Fraud Detection and PaySim Mobile Money as a validation. The pattern of each dataset section is similar: the overall metrics for the three models; metrics by risk tier and a comparison against a simple rule-based baseline. All three are brought together in Section 6.4 and all the results are interpreted in Section 6.5.

6.1 ULB Credit Card Fraud: Primary Dataset

6.1.1 Main Metrics Comparison

The baseline is a plain logistic regression with no deferral (sklearn, random_state = 42) and Models 2 and 3 are the two neural models, both trained with seed 42 (Table 3). The accuracy of both L2D models is over 99.9% with deferral, while Model 1 has a false-alarm problem which is eliminated. The F1 for Model 3 is slightly lower than the F1 for Model 2 (0.738 vs 0.774), which is used in Section 6.1.2 to determine if this is a real cost or seed noise.

The overall deferral rate is not the most important result, which is the distribution across risk tiers (Table 4): Critical-tier deferral increases 4.6x, from 0.21% to 0.96% while Low-tier deferral increases only slightly, from 0.28% to 0.42%, causing deferral to shift toward the tier with the highest potential fraud loss, leaving the low-stakes tier alone.

Table 3: Main metrics comparison across the three models, ULB Credit Card Fraud dataset.

Metric	Model 1 (Baseline)	Model 2 (Standard L2D)	Model 3 (Risk-Sensitive L2D)
F1 Score	0.1094	0.7736	0.7378
ROC-AUC	0.9698	0.9787	0.9800
Accuracy	97.43%	99.92%	99.90%
Deferral Rate	0.00%	0.25%	0.40%
Frauds Missed (of 98)	8	15	14
Frauds Deferred	0	1	1

Table 4: Deferral rate by risk tier, Standard vs Risk-Sensitive L2D, ULB.

Risk Tier	Model 2 (Standard L2D)	Model 3 (Risk-Sensitive L2D)	Ratio
Low (€0-22)	0.28%	0.42%	1.5x
Medium (€22-77)	0.18%	0.22%	1.2x
High (€77-500)	0.25%	0.49%	2.0x
Critical (>€500)	0.21%	0.96%	4.6x

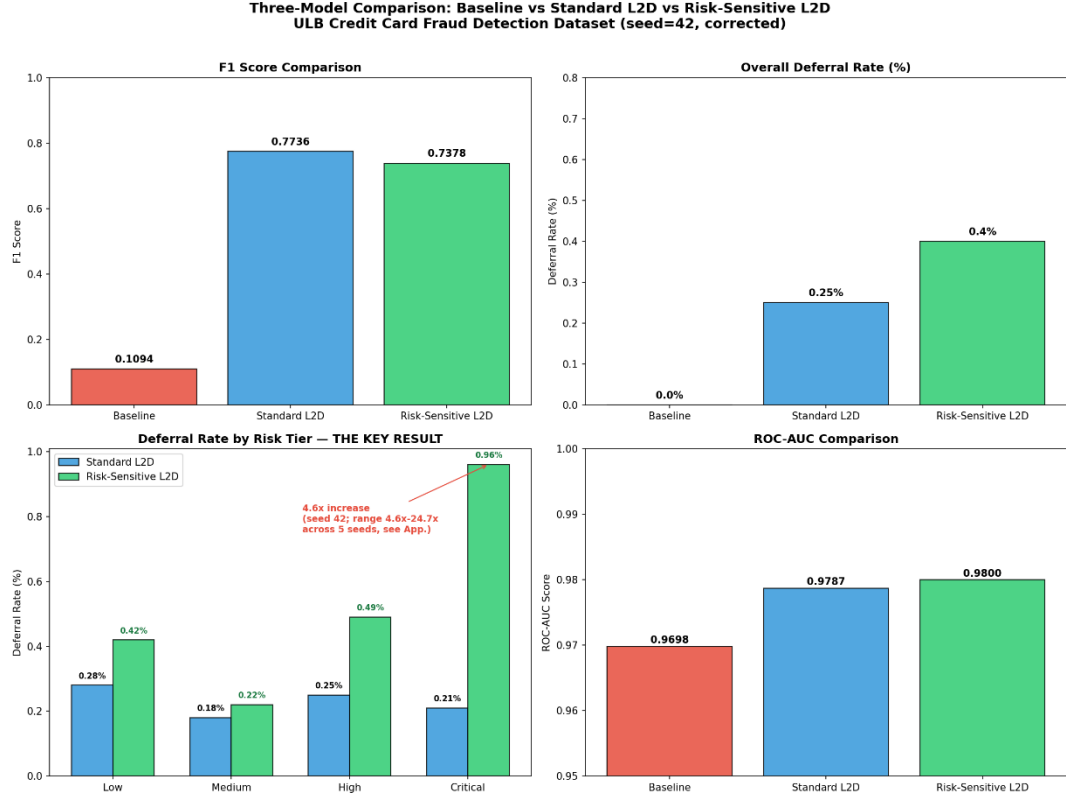


Figure 3: The three-model comparison on ULB based on F1, overall deferral rate, deferral by risk tier and ROC-AUC.

6.1.2 Robustness Checks

Weight initialisation and batch order were random; checked against four other seeds (43-46) and split fixed at seed 42. The number of increases ranged from 4.6x to 24.7x (mean 15.6x) in the direction held up in all five runs. With the shared-trunk head and ULB's extreme imbalance (0.17% fraud) with SMOTE variance, complete inability to classify any fraud case happened in 2 of 5 seeds (44, 46), whereas no case of complete inability happened on any IEEE-CIS seed, which has much more signal (209 features, 3.5% fraud). Tier-ordering was also observed in collapsed runs, which were reported and not discarded simply because they are inconvenient - discarding inconvenient seeds is not robustness. Seed 42 is not the best, it is the pre-registered number one seed - all five seeds with collapse status are reported in Table 5.

The pattern has a reasonably steep cost gradient (Low 1.0, Medium 0.8, High 0.5, Critical 0.2) which is one of several options, a wide spread increased the critical/low ratio to 35.8x, a moderate spread (Critical 0.5+) more or less neutralized it. Only deferral behaviour was affected (F1 0.74-0.78). A one-line rule, defer only if Critical-tier, catches slightly fewer frauds than the learned model (11 vs 14) but sends over eight times more transactions to a human (3.29% vs 0.40%) and the learned head also reaches a higher F1 on the cases it keeps autonomous (0.74 vs 0.57).

Table 5: Critical-tier deferral rate by seed and classifier-collapse status, ULB.

Seed	Model 2 Critical (%)	Model 3 Critical (%)	Critical/Low Ratio (M3)	Classifier Collapsed
42	0.21	0.96	2.29x	No
43	0.53	13.09	25.17x	No
44	2.08	12.82	14.91x	Yes

45	0.59	13.89	29.55x	No
46	0.80	15.28	15.75x	Yes

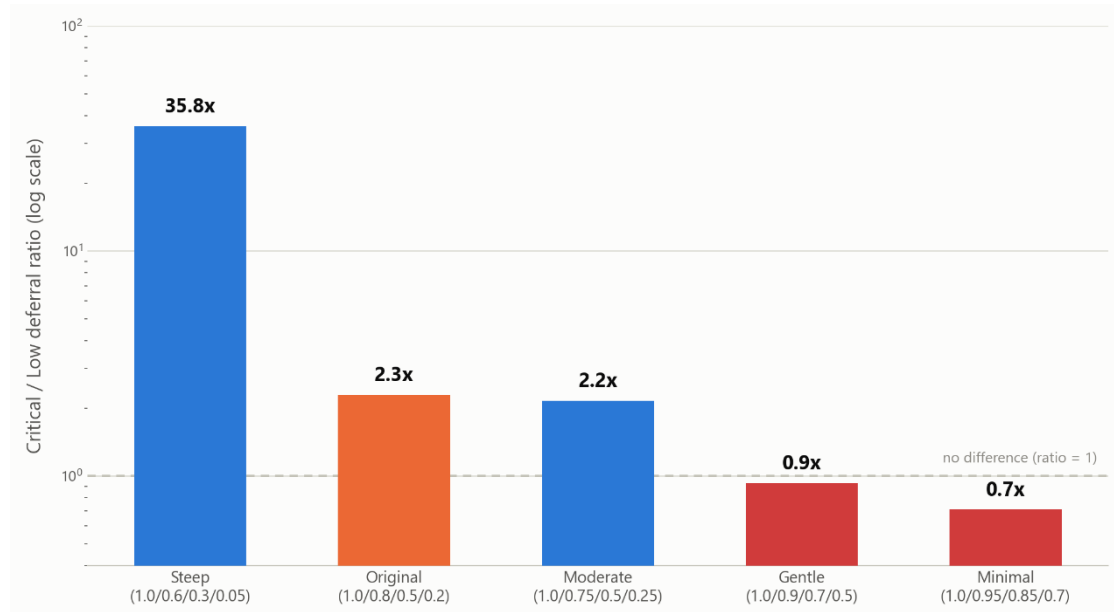


Figure 4: Critical/Low deferral ratio across five risk-weight configurations, ULB.

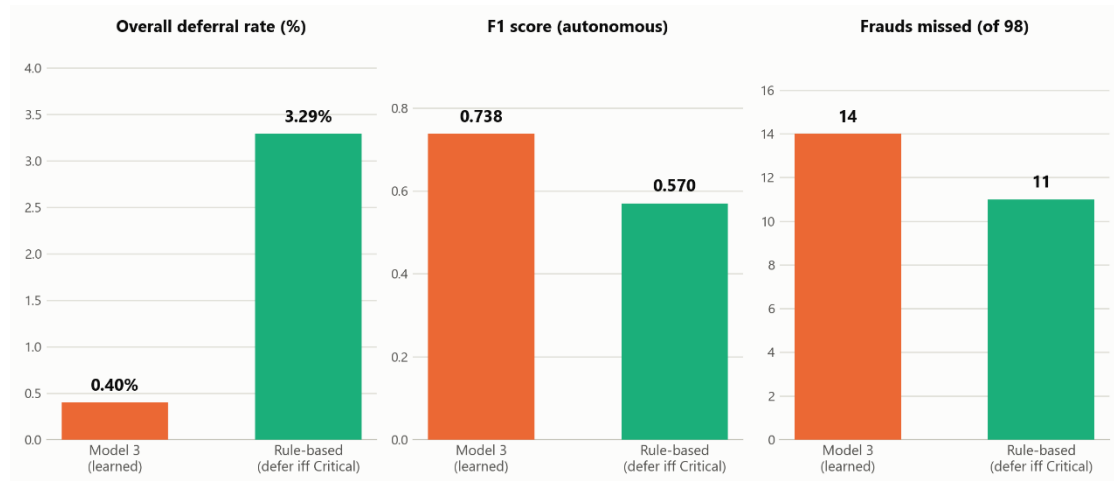


Figure 5: Model 3 (learned deferral) vs a rule-based baseline using the same classifier, ULB.

6.1.3 Statistical Significance

A bootstrap 95% confidence interval (10,000 resamples) on the critical-tier deferral rate difference between Model 2 and Model 3 across the five ULB seeds gives a mean increase of 10.38 percentage points, 95% CI [5.12, 13.68], which excludes zero and indicates the increase is unlikely to be a seed-sampling artefact.

6.2 IEEE-CIS Fraud Detection: Validation Dataset 1

6.2.1 Main Results

The base deferral cost for Model 2 and Model 3 is 0.63 but risk weights are adjusted for this data set; Low 1.0, Medium 0.9, High 0.7, Critical 0.5 (Table 6). It is not a trade-off of F1 here, it's an improvement in F1 (0.400 to 0.559) and ROC-AUC (0.913 to 0.944) and reduction of

missed frauds (903 to 460). The small F1 cost that can be seen on ULB is not a function of the method. In standard L2D, deferral rises in the opposite way, with more on the Low-tier (12.74%) than on the Critical-tier (10.67%) (Table 7), which is corrected in Risk-Sensitive L2D, here the deferral rate increases step-by-step with tier, 18.6%, 13.6%, 41.9%, 70.6%, which is a 6.6x increase in deferral over standard L2D on the Critical-tier.

Table 6: Main metrics comparison across the three models, IEEE-CIS Fraud Detection dataset.

Metric	Model 1 (Baseline)	Model 2 (Standard L2D)	Model 3 (Risk-Sensitive L2D)
F1 Score	0.343	0.400	0.559
ROC-AUC	0.905	0.913	0.944
Deferral Rate	0.00%	8.91%	36.10%
Frauds Missed (of 4,133)	1,030	903	460
Frauds Deferred	0	307	1,437

Table 7: Deferral rate by risk tier, Standard vs Risk-Sensitive L2D, IEEE-CIS.

Risk Tier	Model 2 (Standard L2D)	Model 3 (Risk-Sensitive L2D)
Low (\$0-43)	12.74%	18.60%
Medium (\$43-69)	3.60%	13.62%
High (\$69-125)	8.68%	41.94%
Critical (>\$125)	10.67%	70.56%

**Three-Model Comparison: Baseline vs Standard L2D vs Risk-Sensitive L2D
IEEE-CIS Fraud Detection Dataset (base cost 0.63, harmonised)**

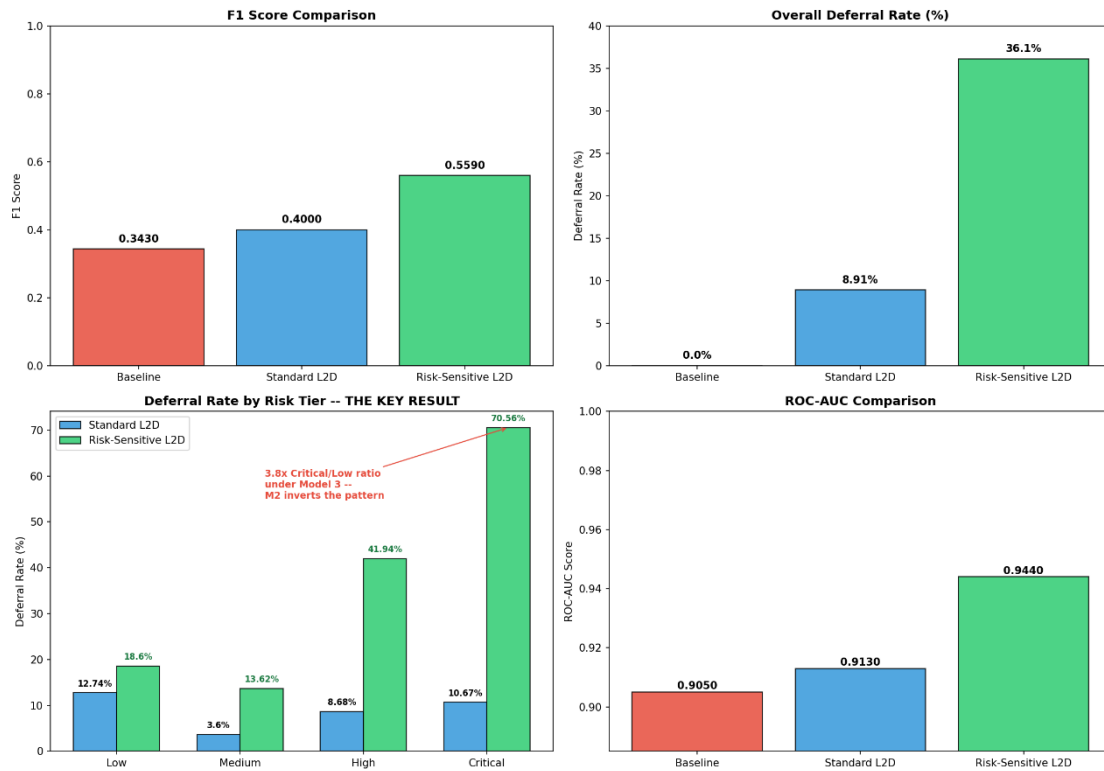


Figure 6: Three-model comparison on IEEE-CIS: F1, overall deferral rate, deferral by risk tier and ROC-AUC.

6.2.2 Robustness Checks

This is a multi-seed check across seeds 42-45, which gives a consistent result, all are between 5.49x and 6.78x, which have mean of around 6.2x, with no collapse of the classifiers. The 209 features from IEEE-CIS and the 3.5% fraud rate provide more signal than ULB's 30 features and just 0.17% fraud rate. A 95% CI, based on 10,000 resamples and on the mean difference of the Critical-tier deferral across the four seeds, is (58.09 points, (56.97 points, 59.25 points) which does not contain 0. Replacing Model 3's learned head with the critical-tier rule is unambiguous here, the rule loses on every measure that matters, with roughly half the F1 (0.29 vs 0.56) and nearly double the missed frauds (820 vs 460) at a broadly similar deferral rate (24.74% vs 36.10%).

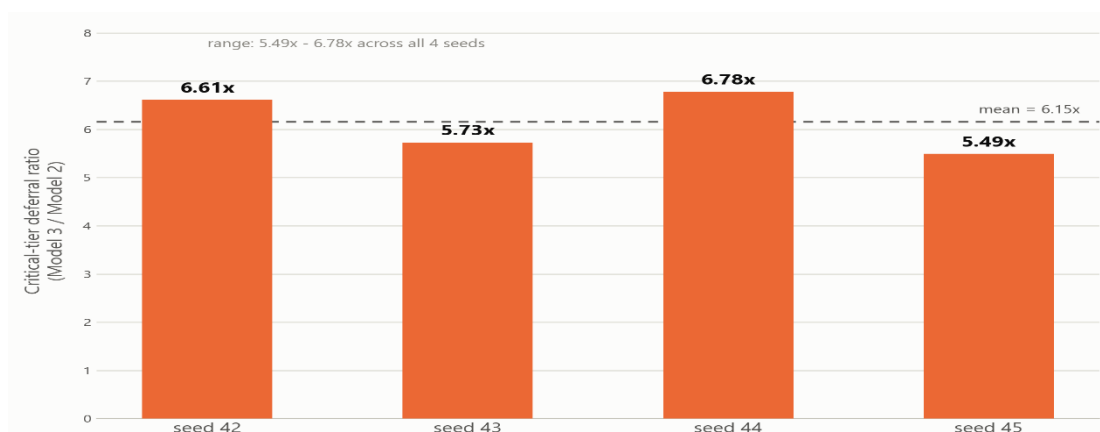


Figure 7: Critical-tier deferral ratio (Model 3 / Model 2) across four training seeds, IEEE-CIS.

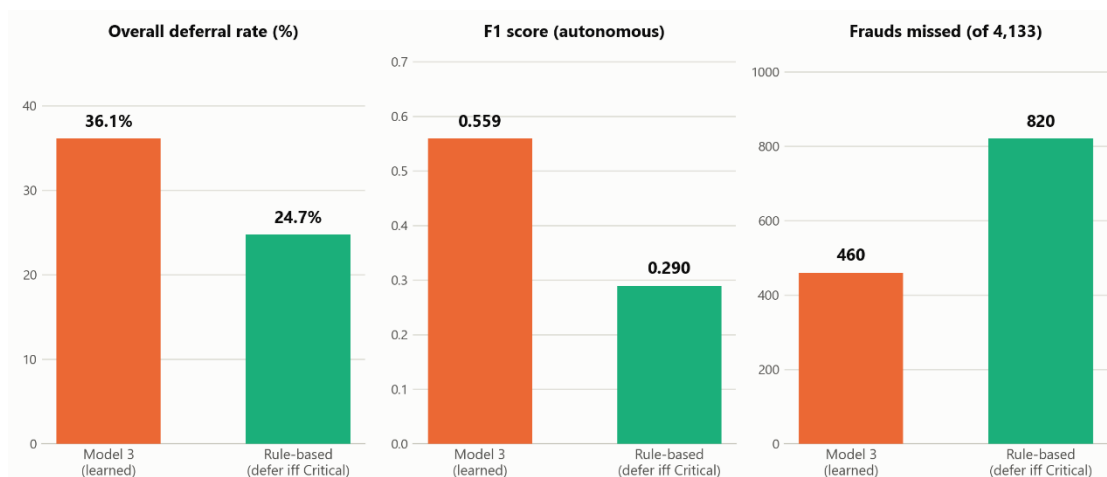


Figure 8: Model 3 (learned deferral) vs a rule-based baseline using the same classifier, IEEE-CIS.

6.3 PaySim Mobile Money: Validation Dataset 2

6.3.1 Main Results

The most challenging of the three was calibrating PaySim. It has only 7 features, so the cost of deferral response is not smooth, with most values of the deferral going to near 0 or near 100% (99.6%). Only the $c_base = 0.4$ (Table 8, Table 9) had been established as a true middle

ground and verified over an entire 15-epoch run, for both Model 2 and Model 3 below. The low deferral rates are not a weakness in the method but rather lack of features. Standard L2D also fails to differentiate between High (0.69%) and Critical (0.68%) again but Risk-Sensitive L2D doubles up on Critical-tier deferral to 3.84%, while Low-tier remains close to zero (0.02%). The scale is smaller but the shape is the same as the other datasets.

Table 8: Main metrics comparison across the three models, PaySim Mobile Money dataset.

Metric	Model 1 (Baseline)	Model 2 (Standard L2D)	Model 3 (Risk-Sensitive L2D)
F1 Score	0.648	0.599	0.617
ROC-AUC	0.998	0.999	0.999
Deferral Rate	0.00%	0.45%	1.28%
Frauds Missed (of 1,643)	13	9	9
Frauds Deferred	0	1	2

Table 9: Deferral rate by risk tier, Standard vs Risk-Sensitive L2D, PaySim.

Risk Tier	Model 2 (Standard L2D)	Model 3 (Risk-Sensitive L2D)
Low (\$0-82,974)	0.00%	0.02%
Medium (\$82,974-171,261)	0.44%	0.51%
High (\$171,261-306,791)	0.69%	0.74%
Critical (>\$306,791)	0.68%	3.84%

**Three-Model Comparison: Baseline vs Standard L2D vs Risk-Sensitive L2D
PaySim Mobile Money Dataset (base cost 0.4, recalibrated)**

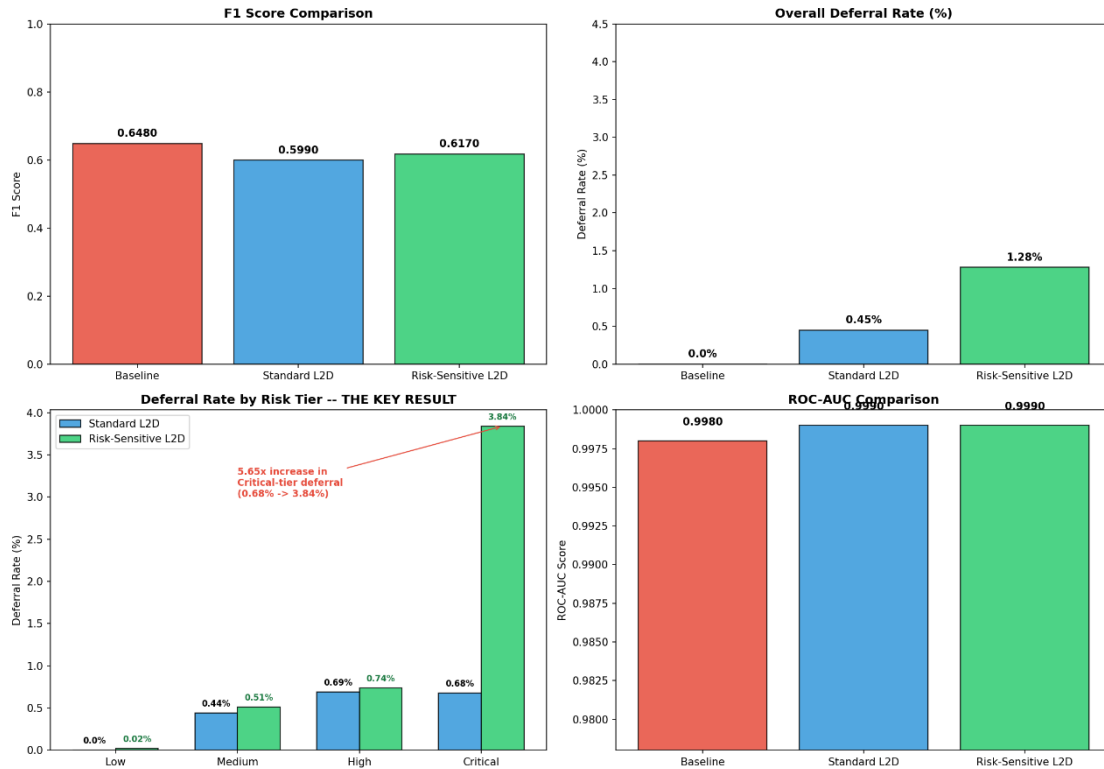


Figure 9: Three-model comparison on PaySim: F1, overall deferral rate, deferral by risk tier and ROC-AUC.

6.3.2 Robustness Checks

The rule-based comparison here is similar to the ULB pattern, as opposed to the IEEE-CIS one, in that frauds missed is close to a tie (8 vs 9) but the rule defers about twenty times as many transactions (25.08% vs 1.28%) and its autonomous-case F1 is significantly lower (0.42 vs 0.62). PaySim dataset was not re-run across seeds because it was already very expensive to find a stable operating point (Section 6.3.1).

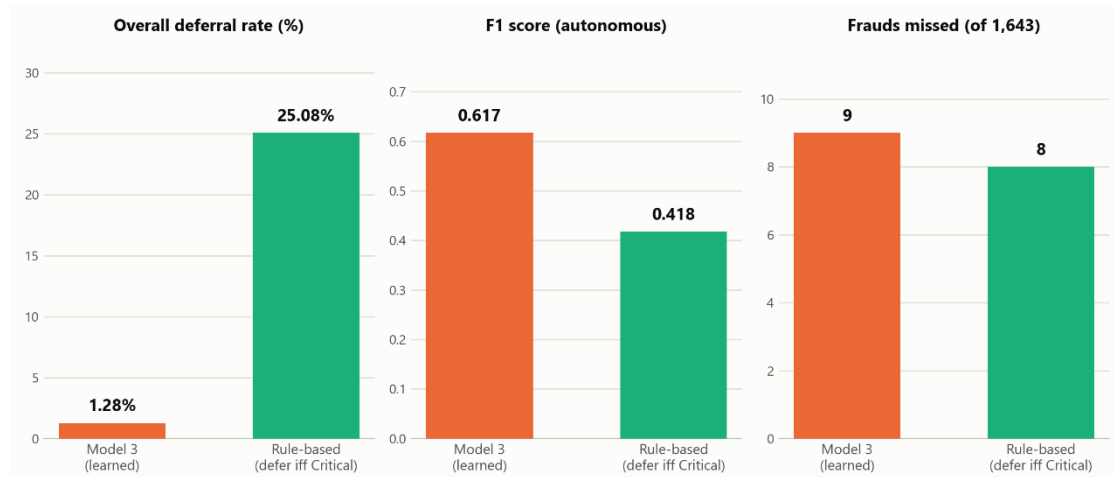


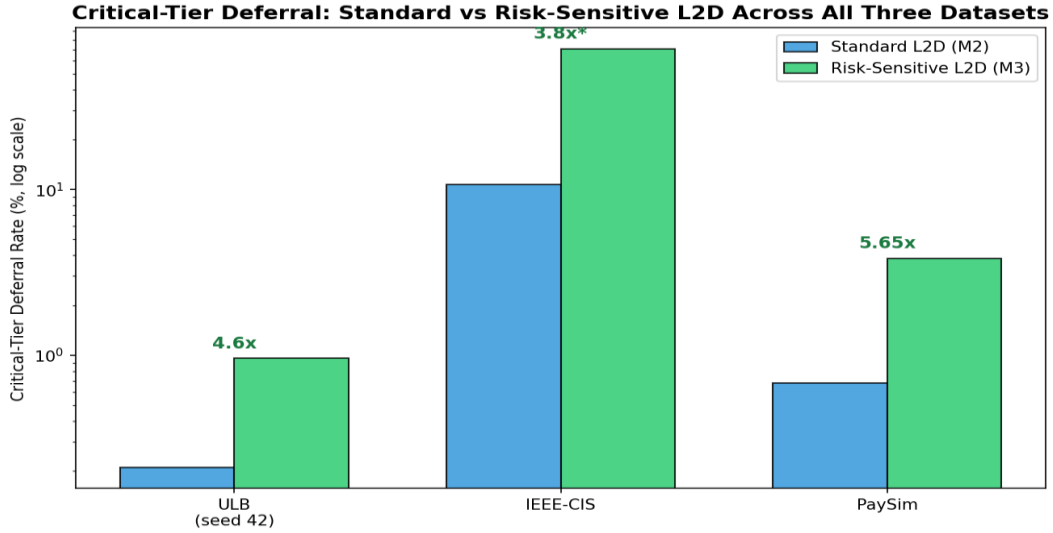
Figure 10: Model 3 (learned deferral) vs a rule-based baseline using the same classifier, PaySim.

6.4 Cross-Dataset Summary

This is true for the three sets of data (Table 10). Standard L2D isn't risk proportional and on IEEE-CIS it's upside down. Risk-Sensitive L2D rectifies this in each and every instance, focusing deferral on the tier where a fraud loss is greatest. The learned mechanism beats a hand-written rule on every measure on IEEE-CIS and is a real contender on raw detection but more efficient on ULB and PaySim, about 8x less deferral for similar number of frauds missed on ULB and 20x less on PaySim.

Table 10: Critical-tier deferral under Standard vs Risk-Sensitive L2D and rule-based comparison, across all three datasets.

Dataset	Critical-tier, M2	Critical-tier, M3	M2 to M3 Ratio	Beats rule-based baseline on raw detection?
ULB	0.21%	0.96%	4.6x (4.6x-24.7x across 5 seeds)	No, efficiency only
IEEE-CIS	10.67%	70.56%	6.6x	Yes, on every metric
PaySim	0.68%	3.84%	5.65x	No, efficiency only



*IEEE-CIS ratio shown is Critical/Low within Model 3 (see Section 6.2.3), not M2-to-M3, since M2 inverts the pattern on this dataset

Figure 11: Critical-tier deferral rate under Standard L2D vs Risk-Sensitive L2D across all three datasets (log scale).

6.5 Discussion

6.5.1 Why Does It Work and Does It Cost Accuracy?

The mechanism is as follows; term of the loss function to be deferred is $w(\text{tier}) * c_{\text{base}} * p_{\text{defer}}$, which is smaller for higher-risk tiers and gradient descent, therefore, prefers to defer the cases with lower weights. What was not known going into the project is whether this is going to hold up with SMOTE oversampling and a shared classification and deferral trunk and the fact that there is a small pool of real Critical-tier fraud cases; the results indicate that it does.

It is a matter of cost, depending on the data set, whether it is accuracy or not. The F1 on ULB is slightly lower due to lesser number of cases to be deferred, leaving more signal for the already small fraud population. On IEEE-CIS the opposite is true: Standard L2D has bad targeting and has hard Critical-tier cases in its own pool and pulls F1 down, while Model 3 has targeted deferral and pulls those same cases. It has no fixed accuracy cost, it is dependent on the degree of the risk mistargeting of the uniform cost baseline. The increase in the deferral of the Critical-tier focuses the review effort where the greatest potential loss is expected; the Critical-tier accounts for 3% of ULB transactions but the highest share of total value of fraud.

$$L_{\text{defer}} = p_{\text{defer}} * w(\text{tier}) * c_{\text{base}}$$

6.5.2 How Robust Is This?

The strength of robustness is a scalar quantity but has no direction. There are five seeds ranging from 4.6x to 24.7x for ULB and the raw output of the deferral head is not well calibrated at 0.5 without calibration, as shown in (Guo *et al.*, 2017). Modern neural networks are not always well calibrated even when training succeeds. IEEE-CIS is much more stable (5.49x to 6.78x, no collapses), probably due to the number of features in it (209) and the amount of fraud in it (3.5%), compared to ULB's 30 features and 0.17%. The same is true of PaySim, which has not been re-verified across seeds but for which a dedicated calibration sweep was required, making it the least well-evidenced of the three (Section 7.2). The ULB sweep indicates that the cost gradient needs to be steep enough to hold and that the headline weights are within that range, although not necessarily at the optimum.

6.5.3 Rule vs Learned Deferral

Can the same number of levels be achieved by a hand-written rule as by a learned mechanism? It varies from one data set to another. A rule deferring all Critical-tier cases is a close second on raw frauds missed for ULB and PaySim, deferring a lot more overall (8x more on ULB, approximately 20x on PaySim). The learned mechanism emerges as a winner on IEEE-CIS with almost double the F1 and half the missed frauds. The difference tracked by dataset signal - with sufficient signal, a learned policy outperforms a fixed rule and with a low signal ratio, a fixed rule is competitive with a learned policy on detection. The constant that the learned mechanism offers is efficiency, which is important considering the capacity constraint in Section 3.4.

6.5.4 Governance and Positioning

Article 14 of the EU AI Act sets out the requirements for human intervention in high-risk systems but offers no guidance on establishing a scalable system for a millions of decisions per day. One method for doing this is to use a different sampling rate on the model, instead of a fixed rate for each model. For both ULB and IEEE-CIS this yields a claim that can be checked, severe-consequence transactions are much more likely to be reviewed by a human being than those that are low-risk.

All the L2D methods reviewed in Chapter 2 are based on an objective that is not aware of the size of the decision being deferred, such as (Mozannar and Sontag, 2020) estimator or the OpenL2D benchmark proposed by (Alves *et al.*, 2025). This thesis substitutes a single instance-dependent cost for a fixed one but does not resolve the practical question from Section 3.4: risk-sensitive deferral only helps if there are enough reviewers for the queue it creates, made concrete by IEEE-CIS's own numbers, where Model 3 defers 70.56% of Critical-tier transactions, a volume whose manageability is beyond this thesis's scope.

7. Conclusion and Future Work

7.1 Restating the Question

The question posed in this thesis was whether a weighting factor in the L2D objective that considers risk will decrease unsafe automation in a high-stakes decision system. The motivation was specific: All published L2D formulations consider low-value and high-value misclassified transactions to be one and the same and none of them had tested the safety of this assumption.

7.2 Answering It

Yes - Risk-sensitive weighting measurably decreases unsafe automation on high-value transactions, correcting a real, reproducible flaw rather than a hypothetical one. Whether in ULB, IEEE-CIS or PaySim, Standard L2D does not defer according to risk - it only distinguishes tiers on ULB with difficulty and it gets the order of deferring tiers on IEEE-CIS wrong and treats High and Critical-tiers as being the same on PaySim. An instance-level risk weight resolves this: a 4.6x increase on ULB (4.6x-24.7x over 5 seeds), 6.6x on IEEE-CIS and a gradual increase on it throughout the tiers, up to 5.65x on PaySim. A pattern with a steep cost gradient was detected by a weight-sensitivity sweep on ULB and a rule-based baseline was performed across all three, leading to the conclusion that the real value of the mechanism is that it is always efficient, rather than an unconditional win (Section 6.5).

That isn't a super easy discovery. Effect size is dependent on initialisation and signal in the datasets, particularly in the case of ULB, it is more variable and in PaySim only checked at one seed. The learned mechanism only fails on a hand-written rule on raw frauds on IEEE-CIS but is only faster and not more accurate on ULB and PaySim, although not in every seed/dataset combination tested. The three primary threats to external validity are a simulated expert signal, seed/dataset-dependent effect sizes and PaySim's single operating point, none of which overturn the main finding, why the result is considered to be robust.

7.3 Key Contribution

The deferral decision is best interpreted as a principled extension of cost-sensitive learning (Elkan, 2001) to the deferral decision, for the first time in the context of L2D and not just plain classification. It is narrow but very broad: not a new estimator or architecture, it is merely one line of loss change, $\text{weight}(\text{tier}) * \text{base_cost}$, that can be plugged into any existing implementation of L2D without redoing all the theory behind it.

7.4 Future Work

There appear to be three directions that need to be explored. First, using real human-expert annotations instead of the simulated deferral signal, to see if the learned boundary is equal to the human-expert's deferral boundary. Second, it will be interesting to test the framework on a domain which is not as likely as currency to have a low risk, such as healthcare triage and examine whether the risk-matrix concept extends beyond currency. Third, the capacity of reviewers is a hard constraint on the model, which could be extended naturally to let the model adapt to a desired deferral rate, with c_base treated as a Lagrange multiplier while keeping the capacity constraint fixed.

It's still quite small but it's still new: ONE line loss change is easier to commercialise than new architecture. The natural next step would be diagnostic - if an organisation is operating an L2D-style system, it could upload its decisions and it would be given a report as to whether their review policy is risk-proportional or not in the same tier-breakdown format as section 6. Otherwise, it would be possible for it to create its own percentile-based risk matrix and size it to the amount of reviewers it has, training only the deferral head, as institutions are not going to replace a system they are working with with one that they have never tried before.

References

- Alves, J.V. *et al.* (2025) “A benchmarking framework and dataset for learning to defer in human-AI decision-making,” *Scientific Data*, 12(1), p. 506. Available at: <https://doi.org/10.1038/s41597-025-04664-y>.
- Arrieta, A.B. *et al.* (2020) “Explainable Artificial Intelligence (XAI): Concepts, Taxonomies, Opportunities and Challenges toward Responsible AI,” *Information Fusion*, 58, pp. 82–115. Available at: <https://doi.org/10.1016/j.inffus.2019.12.012>.
- Bahnsen, A.C., Aouada, D. and Ottersten, B. (2015) “Ensemble of Example-Dependent Cost-Sensitive Decision Trees.” arXiv. Available at: <https://doi.org/10.48550/arXiv.1505.04637>.
- Bansal, G. *et al.* (2021) “Does the Whole Exceed its Parts? The Effect of AI Explanations on Complementary Team Performance,” *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems. CHI '21: CHI Conference on Human Factors in Computing Systems*, Yokohama Japan: ACM, pp. 1–16. Available at: <https://doi.org/10.1145/3411764.3445717>.
- Buçinca, Z., Malaya, M.B. and Gajos, K.Z. (2021) “To Trust or to Think: Cognitive Forcing Functions Can Reduce Overreliance on AI in AI-assisted Decision-making,” *Proceedings of the ACM on Human-Computer Interaction*, 5(CSCW1), pp. 1–21. Available at: <https://doi.org/10.1145/3449287>.
- Buscemi, A. *et al.* (2025) “Assessing High-Risk AI Systems under the EU AI Act: From Legal Requirements to Technical Verification.” arXiv. Available at: <https://doi.org/10.48550/arXiv.2512.13907>.
- Chawla, N.V. *et al.* (2002) “SMOTE: Synthetic Minority Over-sampling Technique,” *Journal of Artificial Intelligence Research*, 16, pp. 321–357. Available at: <https://doi.org/10.1613/jair.953>.
- Dal Pozzolo, A. *et al.* (2015) “Calibrating Probability with Undersampling for Unbalanced Classification,” *2015 IEEE Symposium Series on Computational Intelligence. 2015 IEEE Symposium Series on Computational Intelligence (SSCI)*, Cape Town, South Africa: IEEE, pp. 159–166. Available at: <https://doi.org/10.1109/SSCI.2015.33>.
- Elkan, C. (2001) “The Foundations of Cost-Sensitive Learning,” *Proceedings of the 17th International Joint Conference on Artificial Intelligence*, pp. 973–978.
- Guo, C. *et al.* (2017) “On Calibration of Modern Neural Networks,” *Proceedings of the 34th International Conference on Machine Learning*, PMLR 70, pp. 1321–1330.
- Hemmer, P. *et al.* (2023) “Learning to Defer with Limited Expert Predictions,” *Proceedings of the AAAI Conference on Artificial Intelligence*, 37(5), pp. 6002–6011. Available at: <https://doi.org/10.1609/aaai.v37i5.25742>.
- High-Level Expert Group on Artificial Intelligence (2019) *Ethics Guidelines for Trustworthy AI*. Brussels: European Commission.
- IEEE-CIS / Vesta Corporation (2019) *IEEE-CIS Fraud Detection*. Kaggle. Available at: <https://www.kaggle.com/c/ieee-fraud-detection> (Accessed: 8 July 2026).

Keswani, V., Lease, M. and Kenthapadi, K. (2021) “Towards Unbiased and Accurate Deferral to Multiple Experts,” *Proceedings of the 2021 AAAI/ACM Conference on AI, Ethics, and Society*. Available at: <https://doi.org/10.1145/3461702.3462516>.

Liu, J., Gallego, B. and Barbieri, S. (2022) “Incorporating uncertainty in learning to defer algorithms for safe computer-aided diagnosis,” *Scientific Reports*, 12(1), p. 1762. Available at: <https://doi.org/10.1038/s41598-022-05725-7>.

Lopez-Rojas, E.A., Elmir, A. and Axelsson, S. (2016) “PaySim: A Financial Mobile Money Simulator for Fraud Detection,” *Proceedings of the 28th European Modeling and Simulation Symposium (EMSS 2016)*, Larnaca, Cyprus.

Madras, D., Pitassi, T. and Zemel, R. (2018) “Predict Responsibly: Improving Fairness and Accuracy by Learning to Defer,” *Advances in Neural Information Processing Systems 31 (NeurIPS 2018)*.

Mökander, J. and Floridi, L. (2023) “Operationalising AI governance through ethics-based auditing: an industry case study,” *AI and Ethics*, 3(2), pp. 451–468. Available at: <https://doi.org/10.1007/s43681-022-00171-7>.

Mosqueira-Rey, E. *et al.* (2023) “Human-in-the-loop machine learning: a state of the art,” *Artificial Intelligence Review*, 56(4), pp. 3005–3054. Available at: <https://doi.org/10.1007/s10462-022-10246-w>.

Mozannar, H. *et al.* (2023) “Who Should Predict? Exact Algorithms for Learning to Defer to Humans,” *Proceedings of the 26th International Conference on Artificial Intelligence and Statistics*, PMLR 206.

Mozannar, H. and Sontag, D. (2020) “Consistent Estimators for Learning to Defer to an Expert,” *Proceedings of the 37th International Conference on Machine Learning*, PMLR 119, pp. 7076–7087.

Okati, N., De, A. and Gomez-Rodriguez, M. (2021) “Differentiable Learning Under Triage,” *Advances in Neural Information Processing Systems 34*, pp. 9140–9151.

Pugnana, A. and Ruggieri, S. (2023) “A Model-Agnostic Heuristics for Selective Classification,” *Proceedings of the AAAI Conference on Artificial Intelligence*, 37(8), pp. 9461–9469. Available at: <https://doi.org/10.1609/aaai.v37i8.26133>.

Raji, I.D. *et al.* (2020) “Closing the AI accountability gap: defining an end-to-end framework for internal algorithmic auditing,” *Proceedings of the 2020 Conference on Fairness, Accountability, and Transparency. FAT* '20: Conference on Fairness, Accountability, and Transparency*, Barcelona Spain: ACM, pp. 33–44. Available at: <https://doi.org/10.1145/3351095.3372873>.

“Regulation (EU) 2024/1689 of the European Parliament and of the Council of 13 June 2024 laying down harmonised rules on artificial intelligence (Artificial Intelligence Act)” (2024). Available at: <http://data.europa.eu/eli/reg/2024/1689/oj>.

Sterz, S. *et al.* (2024) “On the Quest for Effectiveness in Human Oversight: Interdisciplinary Perspectives,” *The 2024 ACM Conference on Fairness Accountability and Transparency*, pp. 2495–2507. Available at: <https://doi.org/10.1145/3630106.3659051>.

Xia, G. and Bouganis, C.-S. (2024) “Augmenting the Softmax with Additional Confidence Scores for Improved Selective Classification with Out-of-Distribution Data,” *International Journal of Computer Vision*, 132(9), pp. 3714–3752. Available at: <https://doi.org/10.1007/s11263-024-02029-3>.